

Assignment 7: GLMs (Linear Regressios, ANOVA, & t-tests)

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OVERVIEW

This exercise accompanies the lessons in Environmental Data Analytics on generalized linear models.

Directions

1. Rename this file `<FirstLast>_A07_GLMs.Rmd` (replacing `<FirstLast>` with your first and last name).
2. Change “Student Name” on line 3 (above) with your name.
3. Work through the steps, **creating code and output** that fulfill each instruction.
4. Be sure to **answer the questions** in this assignment document.
5. When you have completed the assignment, **Knit** the text and code into a single PDF file.

!! AI generated code should NOT be used in this assignment

Set up your session

1. Set up your session. Check your working directory. Load the tidyverse, agricolae and other needed packages. Import the *raw* NTL-LTER raw data file for chemistry/physics (NTL-LTER_Lake_ChemistryPhysics_Raw.csv). Set date columns to date objects.
2. Build a ggplot theme and set it as your default theme.

```
#1
#Loading packages
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.1.4      v readr      2.1.5
## v forcats    1.0.0      v stringr   1.5.1
## v ggplot2    4.0.0      v tibble    3.2.1
## v lubridate  1.9.4      v tidyr     1.3.1
## v purrr      1.0.2
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(lubridate)
library(agricolae)
library(here)
```

```
## here() starts at /home/guest/EDE_Fall2025
```

```
library(corrplot)
```

```
## corrplot 0.95 loaded
```

```
#Checking WD
getwd()
```

```
## [1] "/home/guest/EDE_Fall2025"
```

```
here()
```

```
## [1] "/home/guest/EDE_Fall2025"
```

```
#Importing data and correcting dates
Lake <- read.csv(here("./Data/Raw/NTL-LTER_Lake_ChemistryPhysics_Raw.csv"),
                 stringsAsFactors = TRUE)

#Converting dates to date format
Lake$sampldate <- mdy(Lake$sampldate)

#2
#Building and setting theme
a7theme <- theme_classic(base_size = 14)+
  theme(
    axis.title = element_text(color = "darkblue"),
    panel.background = element_rect(fill = "gray"),
    legend.position = "top")

theme_set(a7theme)
```

Simple regression

Our first research question is: Does mean lake temperature recorded during July change with depth across all lakes?

3. State the null and alternative hypotheses for this question:
> Answer: H0: Mean lake temperature recorded during July does not change with depth across all lakes. Ha: Mean lake temperature recorded during July does change with depth across all lakes.
4. Wrangle your NTL-LTER dataset with a pipe function so that the records meet the following criteria:
 - Only dates in July.
 - Only the columns: lakename, year4, daynum, depth, temperature_C

- Only complete cases (i.e., remove NAs)
5. Visualize the relationship among the two continuous variables with a scatter plot of temperature by depth. Add a smoothed line showing the linear model, and limit temperature values from 0 to 35 °C. Make this plot look pretty and easy to read.

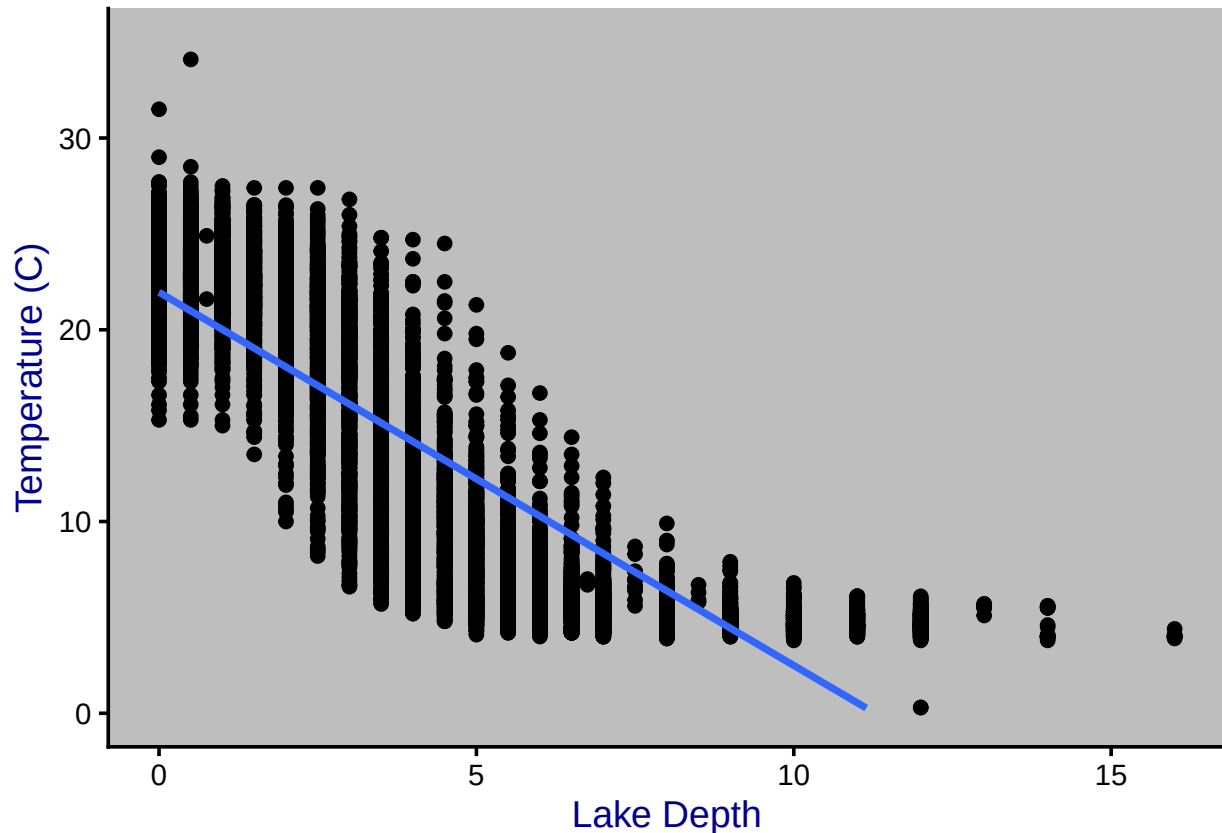
```
#4
Lake.wrangled <- Lake %>%
  mutate(Month = month(sampledate)) %>%
  filter(Month == "7") %>%
  select(lakename, year4, daynum, depth, temperature_C) %>%
  na.omit()

#5
tempvsdepth <- ggplot(Lake.wrangled,
  aes(x=depth,
      y=temperature_C)) +
  geom_point() +
  geom_smooth(method = lm) +
  ylim(0,35) +
  ylab("Temperature (C)") +
  xlab("Lake Depth")

print(tempvsdepth)
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```

```
## Warning: Removed 24 rows containing missing values or values outside the scale range
## ('geom_smooth()').
```



6. Interpret the figure. What does it suggest with regards to the response of temperature to depth? Do the distribution of points suggest about anything about the linearity of this trend?

Answer: The figure suggests a negative correlation between lake depth and temperature. So, as lake depth increases, temperature goes down. However, there is a fairly wide distribution of points, especially at smaller depth values. Given this wide distribution, we might inquire about the strength of the relationship between the two variables. This distribution does taper off after depth values of 8m or so, and there is something of a plateau, suggesting lake temps are consistently low past a certain depth. At this point, they are no longer fitting well with the linear plot.

7. Perform a linear regression to test the relationship and display the results.

```
#7
tempvsdepth.regression <- lm(data = Lake.wrangled, temperature_C~depth)
summary(tempvsdepth.regression)

##
## Call:
## lm(formula = temperature_C ~ depth, data = Lake.wrangled)
##
## Residuals:
```

```
##      Min      1Q  Median      3Q      Max
## -9.5173 -3.0192  0.0633  2.9365 13.5834
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 21.95597    0.06792   323.3  <2e-16 ***
## depth      -1.94621    0.01174  -165.8  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.835 on 9726 degrees of freedom
## Multiple R-squared:  0.7387, Adjusted R-squared:  0.7387
## F-statistic: 2.75e+04 on 1 and 9726 DF, p-value: < 2.2e-16
```

8. Interpret your model results in words. Include how much of the variability in temperature is explained by changes in depth, the degrees of freedom on which this finding is based, and the statistical significance of the result. Also mention how much temperature is predicted to change for every 1m change in depth.

Answer: The model reports that there is indeed a statistically significant, negative correlation between lake temperature and depth. According to the model, 73.87% of the variability in temperature is explained by changes in depth, as indicated by the adjusted r-squared value of 0.7387. This is based on 9726 degrees of freedom, which is fairly high, and we have a p value of 2.2e-16, which is well below .05 and thus indicates statistical significance. And as indicated by the depth estimate, for every 1m change in depth, temperature is predicted to drop by -1.95 degrees celsius.

Multiple regression

Let's tackle a similar question from a different approach. Here, we want to explore what might the best set of predictors for lake temperature in July across the monitoring period at the North Temperate Lakes LTER.

9. Run an AIC to determine what set of explanatory variables (year4, daynum, depth) is best suited to predict temperature.
10. Run a multiple regression on the recommended set of variables.

```
#9
lakeAIC <- lm(data = Lake.wrangled,
              temperature_C ~ depth +
                year4 +
                daynum)
step(lakeAIC)

## Start:  AIC=26065.53
## temperature_C ~ depth + year4 + daynum
##
##              Df Sum of Sq    RSS    AIC
## <none>                141687 26066
```

```
## - year4    1          101 141788 26070
## - daynum   1          1237 142924 26148
## - depth    1          404475 546161 39189

##
## Call:
## lm(formula = temperature_C ~ depth + year4 + daynum, data = Lake.wrangled)
##
## Coefficients:
## (Intercept)      depth      year4      daynum
##    -8.57556    -1.94644     0.01134     0.03978
```

```
#10
lake.bestregression <- lm(data = Lake.wrangled,
                           temperature_C~depth +
                           year4 +
                           daynum)
summary(lake.bestregression)
```

```
##
## Call:
## lm(formula = temperature_C ~ depth + year4 + daynum, data = Lake.wrangled)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -9.6536 -3.0000  0.0902  2.9658 13.6123
##
## Coefficients:
##              Estimate Std. Error  t value Pr(>|t|)
## (Intercept) -8.575564   8.630715  -0.994  0.32044
## depth       -1.946437   0.011683 -166.611 < 2e-16 ***
## year4        0.011345   0.004299   2.639  0.00833 **
## daynum       0.039780   0.004317   9.215 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.817 on 9724 degrees of freedom
## Multiple R-squared:  0.7412, Adjusted R-squared:  0.7411
## F-statistic: 9283 on 3 and 9724 DF,  p-value: < 2.2e-16
```

11. What is the final set of explanatory variables that the AIC method suggests we use to predict temperature in our multiple regression? How much of the observed variance does this model explain? Is this an improvement over the model using only depth as the explanatory variable?

Answer: The AIC method suggested that we include depth, year, and day number. Removing any one of these values resulted in a higher AIC value, indicating a worse fit. The multiple regression for all those variables reports that 74.12% of the variance in temperature can be explained by the three variables. This is only a slight improvement over using only depth as an explanatory variable, which was responsible for 73.87% of the variance.

Analysis of Variance

12. Now we want to see whether the different lakes have, on average, different temperatures in the month of July. Run an ANOVA test to complete this analysis. (No need to test assumptions of normality or similar variances.) Create two sets of models: one expressed as an ANOVA models and another expressed as a linear model (as done in our lessons).

#12

#Running anova with aov function

```
Lake.wrangled.anova <- aov(data = Lake.wrangled, temperature_C ~ lakename)
summary(Lake.wrangled.anova)
```

```
##              Df Sum Sq Mean Sq F value Pr(>F)
## lakename      8  21642   2705.2     50 <2e-16 ***
## Residuals    9719 525813     54.1
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

#Running anova with lm function

```
Lake.wrangled.anova2 <- lm(data = Lake.wrangled, temperature_C ~ lakename)
summary(Lake.wrangled.anova2)
```

```
##
## Call:
## lm(formula = temperature_C ~ lakename, data = Lake.wrangled)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -10.769  -6.614  -2.679   7.684  23.832
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    17.6664     0.6501  27.174 < 2e-16 ***
## lakenameCrampton Lake    -2.3145     0.7699  -3.006 0.002653 **
## lakenameEast Long Lake   -7.3987     0.6918 -10.695 < 2e-16 ***
## lakenameHummingbird Lake -6.8931     0.9429  -7.311 2.87e-13 ***
## lakenamePaul Lake       -3.8522     0.6656  -5.788 7.36e-09 ***
## lakenamePeter Lake      -4.3501     0.6645  -6.547 6.17e-11 ***
## lakenameTuesday Lake    -6.5972     0.6769  -9.746 < 2e-16 ***
## lakenameWard Lake       -3.2078     0.9429  -3.402 0.000672 ***
## lakenameWest Long Lake  -6.0878     0.6895  -8.829 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 7.355 on 9719 degrees of freedom
## Multiple R-squared:  0.03953,    Adjusted R-squared:  0.03874
## F-statistic:    50 on 8 and 9719 DF,  p-value: < 2.2e-16
```

13. Is there a significant difference in mean temperature among the lakes? Report your findings.

Answer: Both ANOVA tests using the aov function and lm function report statistically significant differences in means, meaning we reject the null hypothesis of the mean temperature among all

lakes being the same. The aov function reported a p-value of $<2e-16$. The lm function reported that each lake is a various distance away from the intercept, which represents the mean of all the lakes. And each of those differences is statistically significant. However, we do not know how each lake compares to each of the others.

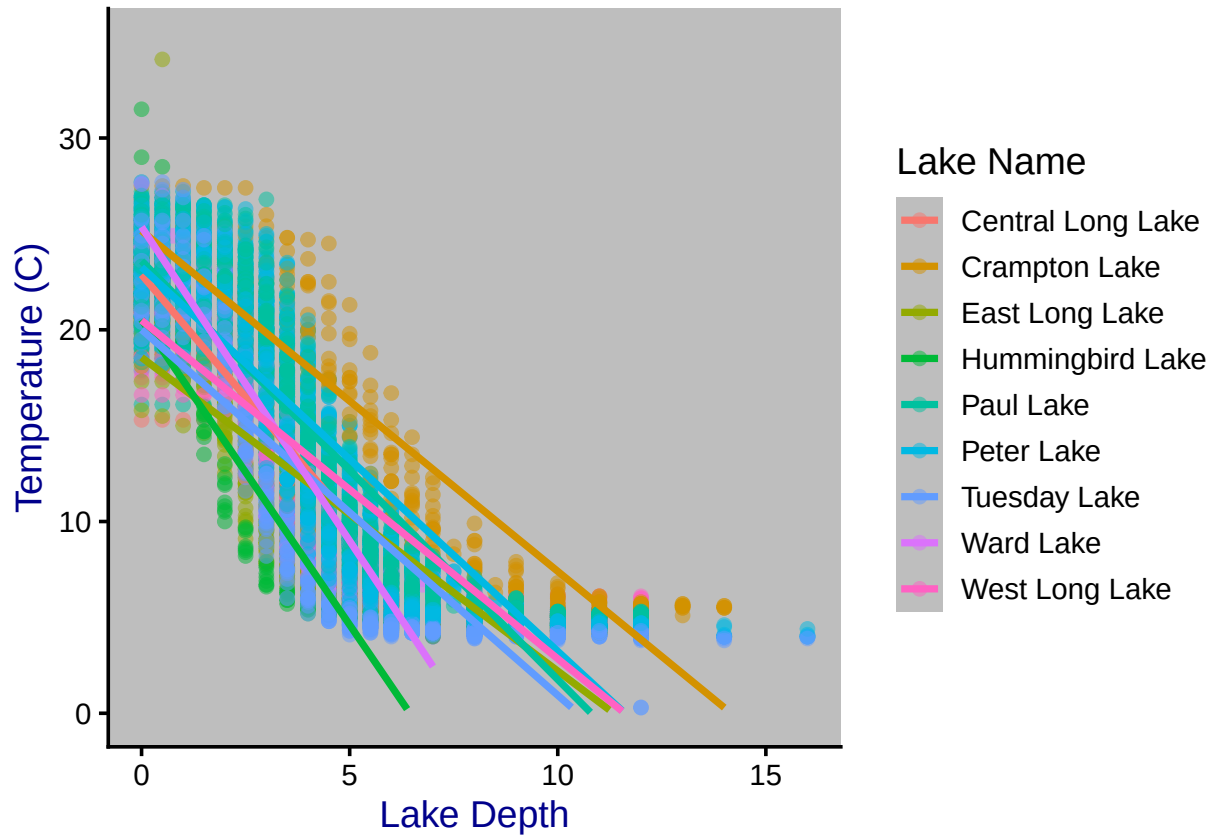
14. Create a graph that depicts temperature by depth, with a separate color for each lake. Add a `geom_smooth` (method = "lm", se = FALSE) for each lake. Make your points 50 % transparent. Adjust your y axis limits to go from 0 to 35 degrees. Clean up your graph to make it pretty.

```
#14. Creating a graph depicting temperature by depth
tempvsdepth.bylake <- ggplot(Lake.wrangled,
                             aes(x=depth,
                                 y=temperature_C,
                                 color=lakename)) +
  geom_point(alpha = 0.5) +
  geom_smooth(method = lm, se = FALSE) +
  ylim(0,35) +
  ylab("Temperature (C)") +
  xlab("Lake Depth") +
  labs(color = "Lake Name")+
  theme(legend.position = "right")

print(tempvsdepth.bylake)
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```

```
## Warning: Removed 73 rows containing missing values or values outside the scale range
## ('geom_smooth()').
```



15. Use the Tukey's HSD test to determine which lakes have different means.

```
#15 Running Tukey HSD
TukeyHSD(Lake.wrangled.anova)
```

```
## Tukey multiple comparisons of means
## 95% family-wise confidence level
##
## Fit: aov(formula = temperature_C ~ lakename, data = Lake.wrangled)
##
## $lakename
##
```

	diff	lwr	upr	p adj
## Crampton Lake-Central Long Lake	-2.3145195	-4.7031913	0.0741524	0.0661566
## East Long Lake-Central Long Lake	-7.3987410	-9.5449411	-5.2525408	0.0000000
## Hummingbird Lake-Central Long Lake	-6.8931304	-9.8184178	-3.9678430	0.0000000
## Paul Lake-Central Long Lake	-3.8521506	-5.9170942	-1.7872070	0.0000003
## Peter Lake-Central Long Lake	-4.3501458	-6.4115874	-2.2887042	0.0000000
## Tuesday Lake-Central Long Lake	-6.5971805	-8.6971605	-4.4972005	0.0000000
## Ward Lake-Central Long Lake	-3.2077856	-6.1330730	-0.2824982	0.0193405
## West Long Lake-Central Long Lake	-6.0877513	-8.2268550	-3.9486475	0.0000000
## East Long Lake-Crampton Lake	-5.0842215	-6.5591700	-3.6092730	0.0000000
## Hummingbird Lake-Crampton Lake	-4.5786109	-7.0538088	-2.1034131	0.0000004
## Paul Lake-Crampton Lake	-1.5376312	-2.8916215	-0.1836408	0.0127491
## Peter Lake-Crampton Lake	-2.0356263	-3.3842699	-0.6869828	0.0000999
## Tuesday Lake-Crampton Lake	-4.2826611	-5.6895065	-2.8758157	0.0000000

```
## Ward Lake-Crampton Lake      -0.8932661 -3.3684639  1.5819317 0.9714459
## West Long Lake-Crampton Lake  -3.7732318 -5.2378351 -2.3086285 0.0000000
## Hummingbird Lake-East Long Lake  0.5056106 -1.7364925  2.7477137 0.9988050
## Paul Lake-East Long Lake      3.5465903  2.6900206  4.4031601 0.0000000
## Peter Lake-East Long Lake     3.0485952  2.2005025  3.8966879 0.0000000
## Tuesday Lake-East Long Lake    0.8015604 -0.1363286  1.7394495 0.1657485
## Ward Lake-East Long Lake      4.1909554  1.9488523  6.4330585 0.0000002
## West Long Lake-East Long Lake  1.3109897  0.2885003  2.3334791 0.0022805
## Paul Lake-Hummingbird Lake    3.0409798  0.8765299  5.2054296 0.0004495
## Peter Lake-Hummingbird Lake   2.5429846  0.3818755  4.7040937 0.0080666
## Tuesday Lake-Hummingbird Lake  0.2959499 -1.9019508  2.4938505 0.9999752
## Ward Lake-Hummingbird Lake    3.6853448  0.6889874  6.6817022 0.0043297
## West Long Lake-Hummingbird Lake 0.8053791 -1.4299320  3.0406903 0.9717297
## Peter Lake-Paul Lake          -0.4979952 -1.1120620  0.1160717 0.2241586
## Tuesday Lake-Paul Lake        -2.7450299 -3.4781416 -2.0119182 0.0000000
## Ward Lake-Paul Lake           0.6443651 -1.5200848  2.8088149 0.9916978
## West Long Lake-Paul Lake      -2.2356007 -3.0742314 -1.3969699 0.0000000
## Tuesday Lake-Peter Lake       -2.2470347 -2.9702236 -1.5238458 0.0000000
## Ward Lake-Peter Lake          1.1423602 -1.0187489  3.3034693 0.7827037
## West Long Lake-Peter Lake     -1.7376055 -2.5675759 -0.9076350 0.0000000
## Ward Lake-Tuesday Lake        3.3893950  1.1914943  5.5872956 0.0000609
## West Long Lake-Tuesday Lake   0.5094292 -0.4121051  1.4309636 0.7374387
## West Long Lake-Ward Lake      -2.8799657 -5.1152769 -0.6446546 0.0021080
```

#Forming groups for easier comparison

```
Lake.wrangled.groups <- HSD.test(Lake.wrangled.anova, "lakename", group = TRUE)
Lake.wrangled.groups
```

```
## $statistics
##      MSerror  Df      Mean      CV
##      54.1016 9719 12.72087 57.82135
##
## $parameters
##      test  name.t ntr StudentizedRange alpha
##      Tukey lakename  9          4.387504 0.05
##
## $means
##               temperature_C      std      r      se Min  Max   Q25   Q50
## Central Long Lake      17.66641 4.196292  128 0.6501298 8.9 26.8 14.400 18.40
## Crampton Lake          15.35189 7.244773  318 0.4124692 5.0 27.5  7.525 16.90
## East Long Lake         10.26767 6.766804  968 0.2364108 4.2 34.1  4.975  6.50
## Hummingbird Lake       10.77328 7.017845  116 0.6829298 4.0 31.5  5.200  7.00
## Paul Lake              13.81426 7.296928 2660 0.1426147 4.7 27.7  6.500 12.40
## Peter Lake             13.31626 7.669758 2872 0.1372501 4.0 27.0  5.600 11.40
## Tuesday Lake           11.06923 7.698687 1524 0.1884137 0.3 27.7  4.400  6.80
## Ward Lake              14.45862 7.409079  116 0.6829298 5.7 27.6  7.200 12.55
## West Long Lake         11.57865 6.980789 1026 0.2296314 4.0 25.7  5.400  8.00
##
##               Q75
## Central Long Lake 21.000
## Crampton Lake    22.300
## East Long Lake    15.925
## Hummingbird Lake 15.625
## Paul Lake         21.400
## Peter Lake        21.500
```

```
## Tuesday Lake      19.400
## Ward Lake         23.200
## West Long Lake    18.800
##
## $comparison
## NULL
##
## $groups
##           temperature_C groups
## Central Long Lake    17.66641    a
## Crampton Lake        15.35189   ab
## Ward Lake            14.45862   bc
## Paul Lake            13.81426    c
## Peter Lake           13.31626    c
## West Long Lake       11.57865    d
## Tuesday Lake         11.06923   de
## Hummingbird Lake     10.77328   de
## East Long Lake       10.26767    e
##
## attr(,"class")
## [1] "group"
```

16. From the findings above, which lakes have the same mean temperature, statistically speaking, as Peter Lake? Does any lake have a mean temperature that is statistically distinct from all the other lakes?

Answer: Based on the Tukey HSD test and resulting groupings, Paul Lake and Ward Lake appear to have the same mean temperature as Peter Lake, statistically speaking, as they are all in group c. Though they do have some differences, their p values are 0.22 and 0.78 respectively, meaning the differences are not statistically significant. It appears that there is no lake whose mean temp is statistically distinct from all the other lakes as each group overlaps with at least one other lake.

17. If we were just looking at Peter Lake and Paul Lake. What's another test we might explore to see whether they have distinct mean temperatures?

Answer: Since we're comparing the means of two samples, we would use a two sample t-test to check if the lakes have distinct mean temperatures. Our null hypothesis for this test would be that the lakes have the same mean temperature and our alternative hypothesis would be that the lakes do not have the same mean temperature.

18. Wrangle the July data to include only records for Crampton Lake and Ward Lake. Run the two-sample T-test on these data to determine whether their July temperature are same or different. What does the test say? Are the mean temperatures for the lakes equal? Does that match you answer for part 16?

```
# Wrangling the data
Crampton.Ward <- Lake.wrangled %>%
  filter(lakename %in% c("Crampton Lake", "Ward Lake"))

# Running two-sample t-test
Crampton.Ward.twosample <- t.test(Crampton.Ward$temperature_C ~ Crampton.Ward$lakename)

Crampton.Ward.twosample
```

```
##
## Welch Two Sample t-test
##
## data: Crampton.Ward$temperature_C by Crampton.Ward$lakename
## t = 1.1181, df = 200.37, p-value = 0.2649
## alternative hypothesis: true difference in means between group Crampton Lake and group Ward Lake is not equal to 0
## 95 percent confidence interval:
## -0.6821129 2.4686451
## sample estimates:
## mean in group Crampton Lake      mean in group Ward Lake
##                15.35189                14.45862
```

Answer: The test results in a p-value of 0.26, which is above .05 and thus not statistically significant. Therefore, we accept the null hypothesis, which states that the means of the two samples are the same / Crampton and Ward Lakes do have the same July temperature. This does match the answer / results of the ANOVA/Tukey HSD, which also demonstrated that Crampton and Ward do not have a statistically significant difference in mean temp.